

Surname

Name

University ID Number

Signature

- =====
- Write the solutions (including procedures and intermediate steps) in the blank areas (use the back of the page, if needed)
 - The number of pages is 4. Additional papers will not be considered.
 - Clarity, order and precision are strongly considered for the final evaluation.
- =====

1. [Analysis of ARMA/ARMAX processes and prediction]

Consider the stochastic process $y(t)$, generated according to the following model

$$y(t) = \frac{1}{2}e(t) + \alpha \cdot e(t-1)$$

$$e \sim wn(0,1).$$

Answer the following questions.

- Compute the covariance function of the process $\gamma_y(\tau)$.
- Considering $\alpha = \frac{1}{2}$, compute the 2-steps ahead optimal predictor of $y(t)$.
- Comment the result, in the light of the previously found covariance function.
- Compute the 2-steps ahead prediction error variance.
- Consider now the 1-step ahead optimal prediction of $y(t)$. Is there a value of α , among the following ones, that minimizes the 1-step ahead prediction error variance?
 - $\alpha = 0$
 - $\alpha = \frac{1}{2}$
 - $\alpha = 2$

Solutions

- The process is an MA(1), so the covariance function should be 0 for $\tau \geq 2$. The covariance function can be computed using its definition, as follows.

$$\begin{aligned}\gamma_y(0) &= \mathbb{E} \left[\left(\frac{1}{2}e(t) + \alpha \cdot e(t-1) \right)^2 \right] = \frac{1}{4} + \alpha^2 \\ \gamma_y(1) &= \mathbb{E} \left[\left(\frac{1}{2}e(t) + \alpha \cdot e(t-1) \right) \left(\frac{1}{2}e(t-1) + \alpha \cdot e(t-2) \right) \right] = \frac{1}{2}\alpha \\ \gamma_y(2) &= \mathbb{E} \left[\left(\frac{1}{2}e(t) + \alpha \cdot e(t-1) \right) \left(\frac{1}{2}e(t-2) + \alpha \cdot e(t-3) \right) \right] = 0 \\ \gamma_y(3) &= \gamma_y(4) = \dots = 0\end{aligned}$$

- The process $y(t)$ must be first written in the Transfer Function form:

$$y(t) = W(z)e(t) = \left[\frac{1}{2} + \frac{1}{2}z^{-1} \right] e(t) \quad \text{with } e(t) \sim wn(0,1)$$

However, $W(z)$ is not in canonical form. First, we notice it is not monic. So, we can re-write the equivalent process.

$$y(t) = \bar{W}(z)\delta(t) = [1 + z^{-1}] \frac{1}{2}e(t) = [1 + z^{-1}]\delta(t) \quad \text{with } \delta(t) \sim wn\left(0, \frac{1}{4}\right)$$

We can see that we have a zero on the unit circle. This would cause problems when trying to compute the predictor from the data as it would not be asymptotically stable. However, writing the process transfer function $\bar{W}(z)$ so to separate the predictable and the unpredictable part:

$$\bar{W}(z) = \frac{C(z)}{A(z)} = E(z) + \frac{R(z)z^{-2}}{A(z)}$$

it results that:

$$\bar{W}(z) = \frac{C(z)}{A(z)} = E(z) + \frac{R(z)z^{-2}}{A(z)} = [1 + z^{-1}] + \frac{0 \cdot z^{-2}}{A(z)}$$

So, the predictable part of the process is, in fact, null. This means that the 2-steps ahead optimal predictor degenerates in the trivial expression:

$$\hat{y}(t + 2|t) = 0$$

In this case, since data are not in fact used, the presence of a zero on the unit circle does not cause any issue.

- c. The covariance function for $\tau = 2$ is equal to zero. This means that there is no correlation between present time data $y(t)$ and 2-steps future data $y(t + 2)$. As a result, the predictor degenerates in the trivial process mean value, which is equal to zero. This confirms the previously found result.

- d. The 2-steps ahead prediction error variance is:

$$\text{var}(\epsilon(t + 2|t)) = \mathbb{E}[(y(t) - \hat{y}(t + 2|t))^2] = \mathbb{E}[(y(t) - 0)^2] = \mathbb{E}[(y(t))^2] = \gamma_y(0) = \frac{1}{2}$$

- e. No, there is no value of α that makes the prediction error variance smaller (which would mean that the process is “more predictable”). In fact, the 1-step optimal prediction error variance is always equal to the noise variance, by construction. Since this variance does not depend on α , for any value the prediction error variance would be 1. Additionally, for $\alpha = \frac{1}{2}$ the one step-ahead predictor is simply stable, thus initial conditions would also affect the prediction error variance.

2. [4SID identification]

Consider a discrete-time dynamic system whose impulse response samples are listed in the following

$$\omega(0) = 0, \omega(1) = 0.5, \omega(2) = 0.1, \omega(3) = 0, \omega(4) = 0, \omega(5) = 0, \omega(6) = 0, \omega(7,8, \dots) = 0$$

Answer the following questions.

- Inspecting the impulse response data, the following system properties can be assessed:
 - The system is (asymptotically stable/unstable)
 - The system features (all/no) (poles/zeros) in $z = 0$
 choose the proper option among the proposed and briefly comment the choice done.
- Estimate the system order from the impulse response data.
- Write a possible state-space system representation.
- Compute the system transfer function and verify the properties selected in point a.
- Consider the following two dynamic systems. Which one is a different representation of the original system?

$$S_1: \begin{cases} x_1(t+1) = u(t) \\ x_2(t+1) = x_1(t) + u(t) \\ y = 0.4 \cdot x_1 + 0.1 \cdot x_2 \end{cases} \quad S_2: \begin{cases} x_1(t+1) = u(t) \\ x_2(t+1) = 2 \cdot x_1(t) \\ y = 0.4 \cdot x_1 + 0.1 \cdot x_2 \end{cases}$$

Solutions

- From the system impulse response one can say that:
 - The system is asymptotically stable, as the impulse response converges to zero.
 - The system features all poles in $z = 0$. In fact, the system impulse response reaches zero in a finite number of steps, thus the system is a Finite Impulse Response system. By definition, these systems feature all poles in $z = 0$.
- The system order can be estimated by building increasing order Hankel matrices, checking the last not null and full rank.

$$\begin{aligned} H_1 &= [\omega(1)] = 0.5 \rightarrow \text{full rank} \\ H_2 &= \begin{bmatrix} \omega(1) & \omega(2) \\ \omega(2) & \omega(3) \end{bmatrix} = \begin{bmatrix} 0.5 & 0.1 \\ 0.1 & 0 \end{bmatrix} \rightarrow \det = -0.01 \rightarrow \text{full rank} \\ H_3 &= \begin{bmatrix} \omega(1) & \omega(2) & \omega(3) \\ \omega(2) & \omega(3) & \omega(4) \\ \omega(3) & \omega(4) & \omega(5) \end{bmatrix} = \begin{bmatrix} 0.5 & 0.1 & 0 \\ 0.1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \det = 0 \rightarrow \text{NOT full rank} \end{aligned}$$

So the system order is $n = 2$.

- The extended $(n + 1)$ Hankel matrix can be factorized as follows:

$$H_3 = \begin{bmatrix} 0.5 & 0.1 & 0 \\ 0.1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = O_3 \cdot R_3 = \begin{bmatrix} 0.5 & 0.1 \\ 0.1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

and the state-space matrix computed, accordingly:

$$\begin{aligned} H &= O_3(1,:) = [0.5 \quad 0.1] \\ G &= R_3(:,1) = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \\ F &= O_3(1:2,:)^{-1} O_3(2:3,:) = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} \end{aligned}$$

- The system transfer function is

$$W(z) = H(zI - F)^{-1}G = \frac{0.5 \cdot z + 0.1}{z^2}$$

which features 2 poles in $z = 0$ and thus is also asymptotically stable, as previously expected.

e. There are several ways to check if the systems are in fact the same, but with a different state space representation.

1. Transfer function

$$W_1(z) = \frac{0.5 \cdot z + 0.1}{z^2} = W(z), W_2(z) = \frac{0.4 \cdot z + 0.2}{z^2} \neq W(z)$$

2. Hankel matrix

$$O_{2,S1} = \begin{bmatrix} 0.4 & 0.1 \\ 0.1 & 0 \end{bmatrix}, R_{2,S1} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \rightarrow H_{2,S1} = \begin{bmatrix} 0.5 & 0.1 \\ 0.1 & 0 \end{bmatrix} = H_2$$

$$O_{2,S2} = \begin{bmatrix} 0.4 & 0.1 \\ 0.2 & 0 \end{bmatrix}, R_{2,S2} = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} \rightarrow H_{2,S2} = \begin{bmatrix} 0.4 & 0.2 \\ 0.2 & 0 \end{bmatrix} \neq H_2$$

Thus, only the system 1 is a different, but equivalent, representation of the original system.

3. [State-Space identification]

Consider the state-space (SS) representation of a discrete-time Linear Time Invariant system. Starting from this representation:

- a) Find the formula to obtain the Transfer-Function representation
- b) Find the formula to obtain the Impulse-Response representation

[See lecture notes](#)

4. [Frequency-domain identification]

Describe in detail the “step 1” (experiment design and pre-processing) of a frequency-domain system identification method.

[See lecture notes](#)